

answer – even if incorrect – has been given for the 'M' mark)

- 6 (a) (i)** resistance is ratio V/I (at a point) **B1**
- either gradient increases or I increases more rapidly than V* **B1 [2]**
- (If states $R = \text{reciprocal of gradient}$, then 0/2 marks here)*
- (ii)** current = 2.00 mA **C1**
- resistance = 2 000 Ω **A1 [2]**
- (b) (i)** straight line from origin **M1**
- passing through (6.0 V, 4.0 mA) (allow $\frac{1}{2}$ square tolerance) **A1 [2]**
- (ii)** individual currents are 0.75 mA and 1/33 mA **C1**
- current in battery = 2.1 mA **A1 [2]**
- (allow argument in terms of $P = I^2 R$ or IV)*
- (c)** same current in R and in C **M1**
- p.d. across C is larger than that across R **M1**
- so since power = VI , greater in C **A1 [3]**
- (allow argument in terms of $P = I^2 R$ or IV)*